

Sustainable use of agricultural residues to improve corn silage quality: Co-ensiling with oregano oil residues

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Highlights

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High-quality silage could be achieved by mixing ensiling corn and ORES.

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ORES suppressed the undesirable fermentation, decreasing ethanol content.

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ORES increased the fungal complexity but not affected bacterial community.

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ORES mitigated the risk of *Paenibacillus* and *Clostridium* contamination.

Abstract

The study aimed to evaluate the effect of [oregano](#) essential oil extraction residues (ORES) on the fermentation quality of [corn silage](#). Non-fungal infected (NFI) and fungal infected (FI) [corn](#) were ensiled with 20% (wet weight) of ORES for 180 d. The inclusion of ORES delayed pH decline regardless of [fungal infection](#). The inclusion of ORES did not adversely affect lactic acid (LA) production in NFI silages but retarded the initial [LA fermentation](#) of FI silages. Co-ensiling [corn](#) with ORES significantly decreased ethanol content. Co-ensiling FI corn with ORES significantly increased the fungal Chao1 and Shannon indices compared to FI-CON silages but did not affect the bacterial Chao1 and Shannon indices. The [fungal infection](#) aggravated the risk of [Paenibacillus](#) and [Clostridium](#) contamination, which was mitigated by co-ensiling with ORES. In conclusion, fungal infection suppressed ensiling fermentation, and heightened the risk of proliferation of undesirable bacteria, while ORES

relieved these risks. Ensiling ORES with corn as forages is a potential and promising strategy for the efficient reusing of agro-industrial waste.

Introduction

Essential oils have emerged as the most promising antimicrobial alternatives due to increasing restrictions on the use of antibiotics worldwide. Oregano essential oil has attracted significant interest from food researchers and manufacturers because of its antioxidant, antibacterial, and antifungal properties. Essential oils are extracted from leaves or shoots via steam distillation, and their yield from the plant biomass is notably low, leading to substantial by-product generation. It is estimated that the production of 1 kg of essential oil results in the generation of approximately 100 kg of solid waste. The trading of essential oils, which exceeds 250,000 tons annually, generates more than 250 million tons of distillation residues (Elena and Jairo René, 2024). These oregano essential oil extraction residues (ORES) are not stable and can undergo quick decay, causing serious concerns for the environment and public health. Although incineration and landfill of these agr-industrial waste have been commonly practiced, they are receiving increasing critiques about environmental sustainability and economic viability (Koul et al., 2022; Ma and Liu, 2019). Utilization and valorization of ORES as a forage for ruminants are feasible and promising because these by-products are rich in bioactive compounds, which showed beneficial biological activities to animals. However, the utilization of ORES as forage is scarcely reported.

Corn silage serves as the primary forage for both beef and dairy cattle, however, it is particularly vulnerable to fungal infections from *Fusarium* and *Aspergillus* species during field growth, leading to contamination with their mycotoxins in corn (da Silva et al., 2020). Especially in recent years, with the global climate change, the increasing frequency of extreme weather events aggravated the risk of fungal diseases in crops in recent years. The fungal infections not only result in yield loss but also compromise the hygienic quality of subsequent corn silages. Queiroz et al. (2012) indicated that rust infestation could predispose silage to poor nutritional value, spoilage, and mycotoxin contamination. Various efforts have been made to suppress various undesirable fermentation and alleviate mycotoxin contamination in silages (Kung et al., 2018a). Chen et al. (2023) reviewed the application of plant essential oils in plant silage fermentation and concluded that essential oils could serve as silage additives to inhibit the proliferation of undesirable microorganisms including *Clostridium*, *Fusarium*, and yeast in silages. Del Valle et al. (2022) demonstrated that Lemongrass essential oils reduced gas losses and butyric acid (BA) concentration in whole-plant sorghum silage. Kung et al. (2008) also demonstrated the

antifungal activities of essential oils toward fungi-producing mycotoxins. However, the use of essential oils in silage production is uneconomic.

Given the broad antimicrobial properties of oregano and the huge amount of ORES after essential oils extraction, ensiling ORES with forage might be a promising avenue. However, the research on the utilization of ORES as forage for ruminants is scarce. The impact of ORES on the silage fermentation remains to be elucidated. This study hypothesized that ORES could inhibit undesirable microbial proliferation without detrimentally impacting the ensiling fermentation. Therefore, the objectives of this study were to determine the effect of ORES on fermentation quality, as well as the microbial communities, of fungal-infected corn silages. It is expected that these may shed light on how to valorize this agr-industrial waste.

Silage preparation and treatments

Corn was cultivated in 8 experimental plots (about 300 m² per plot) at the Nanjing Agricultural University Baima campus (32.04°N, 118.88°E, Nanjing, China). Four plots of corn were artificially infected with mycotoxigenic mold spores successively at silking and kernel blister stage (Wang et al., 2022). The mold spores were prepared as follows: *Aspergillus flavus* and *Fusarium graminearum* were sub-cultured on sterile potato dextrose agar (PDA) medium at 30 °C for 6 d, followed by harvesting

The chemical composition and microbial populations of fresh corn before ensiling

As summarized in Table 1. The fungal infection did not affect ($P > 0.05$) the concentrations of WSC, NDF, ADF, or the populations of LAB and yeasts in fresh corn before ensiling. The fresh FI corn had higher ($P < 0.05$) DM and CP contents, mold count, and numerically ($P > 0.05$) higher NDF and ADF contents than fresh NFI corn before ensiling. This indicated that fungal infection accelerated leaf senescence and lignification of the host plant. Williams et al. (1992) reported that crops severely

Conclusion

This study focused on evaluating the effects of ORES on the preservation characteristics of corn silage, however, it did not pay attention on its effect on nutritional value and feeding characteristics. Therefore, feeding trials should be conducted in future research to evaluate how silages containing ORES influence performance parameters, gastrointestinal traits, and antioxidant capacity in cows, as it is one of the antibiotic alternatives in animal production.

The study demonstrated that the

CRedit authorship contribution statement

Xinbao Liu: Writing – review & editing, Writing – original draft. **Xiaowen Zhang:** Visualization, Validation. **Tianyi Dong:** Supervision, Software. **Yuhong Bao:** Resources, Project administration. **Bin Xu:** Methodology, Investigation. **Yixin Shen:** Funding acquisition, Formal analysis, Data curation, Conceptualization. **Xianjun Yuan:** Conceptualization.

Declaration of competing interest

We would like to submit the enclosed manuscript entitled “**Sustainable utilization of agricultural wastes: a promising method to improve fermentation and hygienic quality of fungal-infected corn silage by co-ensiling with oregano essential oil extraction residues**”, which we wish to be considered for publication in “**Journal of Environmental Management**”.

No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication. I would like to

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